Week 8 Peer Evaluation Report

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DATA 200: Applied Statistical Analysis

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Introduction

As part of the Week 8 deliverable, Team Ghanti Tininini conducted peer evaluations of two projects submitted by classmate groups. Each project is reviewed across five criteria drawn from the course evaluation rubric: dataset and problem definition, exploratory data analysis and preprocessing, statistical modeling and validation, Python application development, and overall presentation and documentation. Scores are given out of 10 per criterion, with a maximum total of 50. Written feedback is provided for each criterion to give constructive and specific comments.

Peer Evaluation 1: Team Paru

Project Overview

Team Paru built a Student Exam Score Predictor using a dataset of 6,607 students with 20 features. Their target variable is exam score (a continuous variable ranging from 55 to 100) and their primary method is Multiple Linear Regression implemented with scikit-learn. The project includes a four-tab Streamlit application covering prediction, feature importance, EDA plots, and project documentation. The team members are Paru Maharjan, Subista Kunwar, and Bipana Khadka.

Criterion 1: Dataset and Problem Definition

Team Paru chose a clearly motivated and practically relevant problem. Predicting student exam performance is a well-established area of educational data mining, and the dataset of 6,607 students with 20 features is appropriately sized for multiple regression. The problem statement is well defined in the application's About tab: predict a continuous exam score from behavioral, socioeconomic, and institutional factors. The distinction between the dependent variable (Exam_Score) and the 20 input features is clearly communicated. The team correctly identified that exam scores above 100 represent data errors and removed them during cleaning, which shows careful thinking about data validity. One area for improvement is that the problem statement in the README and TeamInfo files was left partially blank (the template text was not fully replaced), which slightly weakens the documentation. Overall, the problem is well scoped and the dataset is appropriate.

Table 1

Score: Dataset and Problem Definition

Criterion	Max Score	Score Awarded
Dataset and Problem Definition	**10**	**8**

Criterion 2: Exploratory Data Analysis and Preprocessing

The EDA work is thorough and well structured. The team produced eight output plots covering target distribution with Q-Q check, a correlation heatmap for seven numerical variables, scatter plots for the top two features (Attendance and Hours_Studied), categorical box plots, group mean comparisons, actual vs. predicted, residual diagnostics, and feature importance. This is a comprehensive visual audit of the data. The preprocessing pipeline is clean and documented in the application: rows with Exam_Score above 100 are removed, three columns with missing values (Teacher_Quality, Parental_Education_Level, Distance_from_Home) are imputed using mode, and whitespace is stripped from all string columns. No duplicate records were found.

The ordinal and binary encoding choices are sensible and explicitly mapped in the model training script, with Distance_from_Home correctly inverted so that Near scores higher than Far. Feature selection using a Pearson correlation threshold of absolute r greater than or equal to 0.02

is a reasonable and transparent approach, though a slightly higher threshold might have been worth considering to reduce noise. The Q-Q plot of the exam score distribution is a good addition that many groups skip. Overall, the EDA and preprocessing are well executed.

Table 2

Score: Exploratory Data Analysis and Preprocessing

Criterion	Max Score	Score Awarded
EDA and Preprocessing	**10**	**9**

Criterion 3: Statistical Modeling and Validation

The team built a Multiple Linear Regression model using scikit-learn's LinearRegression with StandardScaler normalization and an 80/20 train-test split. The reported R-squared on the test set is 0.8258, meaning the model explains approximately 82.6% of variance in exam scores. The RMSE of 2.0621 and MAE of 1.6481 are both small relative to the 55-to-100 score range, suggesting strong predictive accuracy. The 5-fold cross-validation R-squared of 0.7358 is notably lower than the test R-squared, which the team correctly acknowledged as mild overfitting in the application's About tab. This self-awareness is commendable.

The diagnostics page runs Shapiro-Wilk on a subsample of 1,000 residuals and reports that residuals are not normal at alpha = 0.05, which is an honest finding that the team does not attempt to hide. Pearson correlation tests and one-way ANOVA tests are also run for Motivation_Level and Family_Income, confirming significant group differences. The standardized coefficient plot makes feature importance easy to interpret and correctly shows Attendance (2.29) and Hours_Studied (1.75) as the dominant positive predictors, with Learning_Disabilities as the only negative predictor. One suggestion is that the gap between CV R-squared and test R-squared warrants more explicit discussion, and the non-normality of residuals should be addressed with a note about its implications rather than simply reported. Nevertheless, the modeling work is strong overall.

Table 3

Score: Statistical Modeling and Validation

Criterion	Max Score	Score Awarded
Statistical Modeling and Validation	**10**	**8**

Criterion 4: Python Application Development

The Streamlit application is well built and clearly the strongest part of this submission. The interface has four tabs: Predictor, Feature Importance, EDA Plots, and About. The Predictor tab is interactive and well designed, with sliders for numerical inputs and dropdowns for categorical ones, all 16 features exposed in a clean four-column layout. The prediction output includes a large color-coded score display (green for Excellent, blue for Very Good, amber for Average, red for Needs Improvement), a progress bar, and a personalized improvement tips section that reads directly from the input values and the coefficient estimates. This tips feature is a particularly thoughtful addition that goes beyond the basic deliverable and gives the application real practical value.

The EDA Plots tab provides a dropdown to view all eight diagnostic outputs. The model is loaded from a pre-trained pickle file at startup, so the interface responds instantly. The application handles all 16 features correctly and clips the prediction to the valid range of 55 to 100. The requirements.txt is present and concise. This is one of the better application implementations in the cohort.

Table 4

Score: Python Application Development

Criterion	Max Score	Score Awarded
Python Application Development	**10**	**10**

Criterion 5: Presentation and Documentation

The application itself serves as the primary documentation vehicle and it does this effectively. The About tab contains a clear project overview, methodology summary, key findings table, and limitations list. The code is well commented and logically structured across five separate files (Home.py and four pages), making it easy to follow the full pipeline from cleaning to diagnostics. The README in the repository was not fully completed (team name and task division fields still contain placeholder template text), which is a minor but noticeable gap in documentation quality.

There are no weekly report documents submitted alongside the code, which limits the depth of written explanation. The coefficient table in the application is accurate and the improvement tips add narrative context to the numerical output. Overall, the documentation embedded in the application is good, but the written project documentation outside the application could be more complete.

Table 5

Score: Presentation and Documentation

Criterion	Max Score	Score Awarded
Presentation and Documentation	**10**	**7**

Summary Score: Team Paru

Table 6

Overall Score Summary

Criterion	Max Score	Score Awarded
Dataset and Problem Definition	**10**	**8**
EDA and Preprocessing	**10**	**9**
Statistical Modeling and Validation	**10**	**8**
Python Application Development	**10**	**10**
Presentation and Documentation	**10**	**7**
TOTAL	**50**	**42 / 50**

Overall Feedback

Team Paru has submitted a strong project with a clearly defined problem, good EDA, a well-performing regression model, and the best application interface of the two projects reviewed. The model achieves an impressive R-squared of 0.8258 on the test set and the Streamlit app is polished, interactive, and genuinely useful. The main areas to improve are: completing the README documentation with actual team information and task division, addressing the gap between cross-validation and test R-squared more explicitly in the discussion, and providing a brief interpretation of the non-normal residual finding rather than simply reporting it. The improvement tips feature in the predictor is a creative addition that shows the team thought carefully about how the model output could be made actionable for end users. Well, done overall.

Peer Evaluation 2: Team Chasma Wizards

Project Overview

Team Chasma Wizards built a Lung Cancer Study application using a dataset of 1,157 patient records with 16 binary and one numerical feature. Their target variable is lung cancer diagnosis (binary: YES/NO) and their primary method is Logistic Regression implemented with scikit-learn inside a StandardScaler pipeline. The project is structured across four Jupyter notebooks covering EDA, feature selection, hypothesis testing, and modeling, plus a single-file Streamlit application with six navigation pages. Team members are Dinesh Karki (Data Engineering and Preprocessing), Kushal Thapa (Modeling and Evaluation), Sanskar Dhakal (Statistical Analysis and Feature Engineering), and Muhammad Sharif Miya (Application Development).

Criterion 1: Dataset and Problem Definition

The problem is well defined and medically relevant. Predicting lung cancer from clinical and lifestyle symptoms is a meaningful binary classification task with real-world implications. The dataset of 1,157 records is smaller than ideal for a machine learning project, but it is appropriate for a logistic regression study and the team clearly understood its limitations. The distinction between the target variable (lung_cancer) and the 15 predictor features is clearly stated in the EDA notebook, which includes a variable overview table classifying each column as numerical independent, dichotomous independent, or dichotomous target.

The team correctly identified that the 1-vs-2 encoding of binary symptoms was non-standard and re-mapped all of them to 0 and 1 during preprocessing. The TeamInfo file is fully completed with actual team member names, roles, and a clear task division, which is a strength compared to some other submissions.

Table 8

Score: Dataset and Problem Definition

Criterion	Max Score	Score Awarded
Dataset and Problem Definition	**10**	**9**

Criterion 2: Exploratory Data Analysis and Preprocessing

The EDA and preprocessing work are solid and shows genuine analytical thinking. The team identified and removed 264 duplicate records (out of 1,157), and importantly they went further than most groups by analyzing the class distribution before and after deduplication. They noted that the duplicate entries disproportionately belonged to the cancer-negative class, introducing bias, and they documented this finding explicitly in the notebook. This is a level of critical thinking about data quality that is uncommon at this stage.

The EDA notebook confirms no missing values were present, which simplified preprocessing. Binary encoding was applied correctly with an explicit mapping dictionary covering all categorical columns, and column names were standardized to lowercase. The Streamlit application extends the EDA with interactive Plotly visualizations including an age distribution histogram with mean marker, symptom prevalence bar charts, a before-versus-after class balance comparison, and a heatmap of symptom rates stratified by cancer status.

The chi-square feature selection analysis in Week 4 is well documented, correctly identifying swallowing difficulty, chest pain, shortness of breath, and smoking as the top four predictors. One minor criticism is that the EDA notebook does not include scatter plots, which were listed as required in the Week 3 deliverable brief, though the application compensates for this partially.

Table 9

Score: Exploratory Data Analysis and Preprocessing

Criterion	Max Score	Score Awarded
EDA and Preprocessing	**10**	**9**

Criterion 3: Statistical Modeling and Validation

The modeling pipeline is well constructed and the results are impressive. The team used backward elimination via logistic regression to reduce features from 15 to 9, dropping peer_pressure, gender, age, yellow_fingers, chronic_disease, and fatigue based on p-values exceeding 0.05. This is a principled feature selection approach and it is executed correctly using statsmodels. The final logistic regression model, trained on a 70/30 split with StandardScaler preprocessing, achieves accuracy of 92.34%, precision of 95.95%, recall of 89.47%, F1-score of 92.59%, and an ROC-AUC of 0.9852.

The AUC of 0.9852 is outstanding for a dataset of this size and reflects a very clean separation between the two classes. The confusion matrix shows 125 true negatives, 128 true positives, 6 false positives, and 15 false negatives. In a medical screening context, false negatives (missing actual cancer cases) are more costly than false positives, so a recall of 89.47% is a reasonable outcome.

The chi-square test with feature tier classification (strong, moderate, weak) is a good addition. The hypothesis testing notebook clearly states the null and alternative hypotheses and correctly concludes that the null is rejected. However, the team did not report cross-validation results, which would have been valuable for assessing overfitting given the relatively small sample of 911 records after deduplication. This is the main gap in the modeling work.

Table 10

Score: Statistical Modeling and Validation

Criterion	Max Score	Score Awarded
Statistical Modeling and Validation	**10**	**9**

Criterion 4: Python Application Development

The Streamlit application is the strongest aspect of this project and is the best-designed interface of the two projects reviewed. The application has six navigation pages: Original Dataset, Cleaned Dataset, EDA, Feature Analysis, Hypothesis, and Model. Each page has a consistent dark-themed design using a custom CSS stylesheet with Plotly charts throughout, metric cards, and color-coded information boxes.

The Feature Analysis page correctly recomputes chi-square scores at runtime and renders an interactive bar chart with tier color coding. The Hypothesis page presents the null and alternative hypothesis statements in a styled card and includes a visual verdict panel. The Model page displays all five metrics as live-computed values from the loaded pipeline, renders the confusion matrix as an interactive Plotly heatmap, and includes a live prediction tool where a user selects symptom values from dropdowns and receives an instant positive or negative classification with a probability gauge. The application trains the model at startup using cached functions, which is architecturally clean. The use of Plotly instead of Matplotlib gives the application a professional visual quality. The only structural gap is that the application expects the data files in the same directory rather than a relative path, which may cause issues on some machines without adjustment.

Table 11

Score: Python Application Development

Criterion	Max Score	Score Awarded
Python Application Development	**10**	**10**

Criterion 5: Presentation and Documentation

The project is well documented across its notebooks and application. Each notebook cell is accompanied by a markdown explanation describing what the code does and why, which is good practice and makes the work accessible to a non-technical audience. The Week 4 notebook includes a summary table classifying features as important, moderate, or insignificant predictors, and the variable overview table in the EDA notebook is a clear and well-structured addition.

The application itself contains informative box annotations on each page that summarize key findings. The README in the TeamInfo file is fully completed with team name, members, and a clear four-way task division. A PDF report (DATA200_Project.pdf) is also included in the repository, which is the only project of the two reviewed to include a standalone written report file.

The main gap is that the project does not include weekly report documents for each week, which would have provided a more complete paper trail of the decision-making process. The PDF report covers the project but does not include the level of statistical detail that weekly deliverables would provide.

Table 12

Score: Presentation and Documentation

Criterion	Max Score	Score Awarded
Presentation and Documentation	**10**	**8**

Summary Score: Team Chasma Wizards

Table 13

Overall Score Summary

Criterion	Max Score	Score Awarded
Dataset and Problem Definition	**10**	**9**
EDA and Preprocessing	**10**	**9**
Statistical Modeling and Validation	**10**	**9**
Python Application Development	**10**	**10**

Presentation and Documentation	**10**	**8**
TOTAL	**50**	**45 / 50**

Overall Feedback

Team Chasma Wizards has submitted an excellent project. The problem definition is clear, the preprocessing is careful and analytically thoughtful (particularly the class imbalance analysis before and after deduplication), and the logistic regression model achieves outstanding discriminative performance with an AUC of 0.9852. The Streamlit application is the most polished of the two projects reviewed, with a custom dark theme, six well-structured pages, interactive Plotly charts throughout, and a live prediction tool with a probability gauge.

The four-member task division is clearly documented and the PDF report is a bonus submission. The main areas to improve are: adding cross-validation reporting to the model evaluation, including scatter plots in the EDA as required by the brief, and addressing the relative data file path issue in the application for easier portability. This is one of the strongest submissions in the cohort.